Lecture 7
Substitutions

We have so far studied 1st-order equations

$$(*) \quad \frac{dy}{dx} = f(x,y), \quad y' + P(x)y = Q(x), \quad M(x,y)dx + N(x,y)dy = 0$$

when they are separable, linear, or exact. In this lecture we will explore other solution techniques that transform a given equation into one of the above forms.

The technique we will use is a substitution to alter the form of our equations:

1. Identify the type of substitution
2. Convert the equation using the substitution into an equation in a different set of variables $x, y \to x, v$ or $x, y \to z$
3. Solve the transformed equation $(*)$
4. Convert back into the original set of variables to express a solution to the original equation

There are many different types of substitutions. We will focus on three different varieties in this lecture:

Definition (Homogeneous Equations)**:** Consider the equation

$$\frac{dy}{dx} = f(x,y). \quad \text{1st order}$$

If $f(x,y)$ can be expressed as a function of (either) $\left(\frac{y}{x}\right)$ (or $\left(\frac{x}{y}\right)$) then the equation is said to be **homogeneous**.

Example: Consider $\frac{dy}{dx} = \frac{y-x}{x} = \frac{y}{x} - 1 = G\left(\frac{y}{x}\right)$

homogeneous

$$F(x,y) = \frac{y-x}{x}$$
$$= \frac{y}{x} - \frac{x}{x} = \frac{y}{x} - 1$$

Example: Is $(x' - 2y' + 1)dx + (x' - y')dy = 0$ homogeneous?

~~[illegible]~~ Divide by x

$$\frac{1}{x}\left((x - 2y + 1)dx + (x - y)dy = 0\right)$$
$$\left(1 - 2\left(\frac{y}{x}\right) + \frac{1}{x}\right)dx + \left(1 - \left(\frac{y}{x}\right)\right)dy = 0$$

not $\frac{y}{x}$ not homogeneous

Example: Is $(x^2 - 2xy)dx + \left(3y^2 - x^2 \sin\left(\frac{x}{y}\right)\right)dy = 0$ homogeneous?

Divide by x^2: $\left(1 - 2\frac{y}{x}\right)dx + \left(3 \cdot \frac{y^2}{x^2} - 1 \cdot \sin\left(\frac{x}{y}\right)\right)dy = 0$

$$\Rightarrow \left(1 - 2 \cdot \frac{y}{x}\right)dx + \left(3 \cdot \left(\frac{y}{x}\right)^2 - \sin\left(\frac{x}{y}\right)\right)dy = 0$$

homogeneous

When we have a homogeneous equation we will use the substitution $v = \frac{y}{x}$ (or $v = \frac{x}{y}$) which will convert the differential equation into a separable equation:

$$\frac{dy}{dx} = f(x,y) \quad \Rightarrow \quad \frac{dy}{dx} = G\left(\frac{y}{x}\right) \quad \Rightarrow \quad \frac{dy}{dx} = G(v) \tag{7.1}$$

using

$$v = \frac{y}{x} \quad \Rightarrow \quad y = xv \quad \Rightarrow \quad \frac{dy}{dx} = \frac{d}{dx}(xv) = 1 \cdot v + x\frac{dv}{dx}$$

new variables are x and v convert $\frac{dy}{dx} \to \frac{dv}{dx}$

substituting into (7.1) we obtain $\frac{dy}{dx} = v + x\frac{dv}{dx}$

$$\frac{dy}{dx} = G(v) \quad \Rightarrow \quad \boxed{v + x\frac{dv}{dx} = G(v)}$$

which is a separable equation.

If we begin with an equation of the form $M(x,y)dx + N(x,y)dy = 0$ we can modify our substitution $\frac{dy}{dx} = v + x\frac{dv}{dx}$ by multiplying by dx and use

$$dy = vdx + xdv$$

all sums of powers in each term are equal: 2

Example: Consider $(x^2 + y^2)dx + (x^2 - xy)dy = 0$.

Divide by x^2: $(1 + \frac{y^2}{x^2})dx + (1 - \frac{y}{x})dy = 0$

$\Rightarrow (1 + (\frac{y}{x})^2)dx + (1 - \frac{y}{x})dy = 0$ homogeneous!

Use $\boxed{v = \frac{y}{x}} \Rightarrow xv = y \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\Rightarrow \boxed{dy = vdx + xdv}$

$\Rightarrow (1 + v^2)dx + (1 - v)\underbrace{(vdx + xdv)}_{dy} = 0$

$\frac{y}{x}$

$\Rightarrow (1 + v^2)dx + vdx - v^2dx + xdv - xvdv = 0$

$\Rightarrow (1 + \cancel{v^2} + v - \cancel{v^2})dx + x(1 - v)dv = 0$

$\Rightarrow (1 + v)dx + x(1 - v)dv = 0$ separable

$\Rightarrow x(1 - v)dv = -(1 + v)dx$

$\Rightarrow (1 - v)dv = -(1 + v)\frac{1}{x}dx$

$\Rightarrow -\frac{1 - v}{1 + v}dv = +\frac{1}{x}dx$

$\Rightarrow \int \frac{v - 1}{v + 1}dv = -\int \frac{1}{x}dx$

$\Rightarrow \int 1 - \frac{2}{v+1}dv = \ln|x| + C$

$v+1 \overline{)\, v - 1}$
$\quad -(v+1)$
$\quad -2$

$\Rightarrow v - 2\ln|v+1| = \ln|x| + C$

reconvert back to x, y $v = \frac{y}{x}$

$\Rightarrow \boxed{\frac{y}{x} - 2\ln\left|\frac{y}{x} + 1\right| = \ln|x| + C}$ implicit

Example: Consider the IVP

$$x\frac{dy}{dx} = y + xe^{y/x}, \quad y(1) = 1$$

← need to reconvert before using

Divide by x

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} + e^{y/x} \quad \text{homogeneous}$$

use: $v = \frac{y}{x} \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$\Rightarrow \left(v + x\frac{dv}{dx}\right) = v + e^{v}$$

$$\Rightarrow x\frac{dv}{dx} = e^{v} \quad \text{separable}$$

$$\Rightarrow \frac{1}{e^{v}}\,dv = \frac{1}{x}\,dx$$

$$\Rightarrow \int e^{-v}\,dv = \int \frac{1}{x}\,dx$$

$$\Rightarrow -e^{-v} = \ln|x| + C$$

$$\Rightarrow e^{-v} = -\ln|x| + C$$

$$\Rightarrow -v = \ln(-\ln|x| + C)$$

$$\Rightarrow v = -\ln(-\ln|x| + C)$$

reconvert $\Rightarrow \frac{y}{x} = -\ln(-\ln|x| + C)$

$$\Rightarrow y = -x\ln(-\ln|x| + C)$$

$y(1) = 1$: $1 = y(1) = -1 \cdot \ln(\underbrace{-\ln(1)}_{0} + C)$

$$\Rightarrow 1 = -\ln(C) \Rightarrow \ln(C) = -1$$

$$\Rightarrow C = e^{-1}$$

So the particular solution is $\boxed{y = -x\ln(-\ln|x| + e^{-1})}$

Example: Suppose we have two functions, $x(t)$ and $y(t)$, of the variable t whose derivatives are related together in terms of a linear function of x and y. The system of their derivatives can be written as

parametric equations

$(x(t), y(t))$

slope $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

$$\begin{cases} \dfrac{dy}{dt} = ax + by, \\ \dfrac{dx}{dt} = cx + dy \end{cases}$$

assumption

We can create an associated 1st-order differential equation by division with

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{ax + by}{cx + dy}$$

homogeneous

which is a homogeneous equation. The solutions to this equation are called **integral curves**.

Find the integral curves of

$$\begin{cases} \dfrac{dy}{dt} = -4x - y \\ \dfrac{dx}{dt} = 2x - y \end{cases}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-4x - y}{2x - y} = \frac{\cancel{x}(-4 - \frac{y}{x})}{\cancel{x}(2 - \frac{y}{x})} = \frac{-4 - \frac{y}{x}}{2 - \frac{y}{x}}$$

factor on x

use $v = \frac{y}{x}$, $\frac{dy}{dx} = v + x\frac{dv}{dx} \Rightarrow v + x\frac{dv}{dx} = \frac{-4 - v}{2 - v}$ separable

$$\Rightarrow x\frac{dv}{dx} = \frac{-4 - v}{2 - v} - v = \frac{-4 \cancel{-v} - (2 \cancel{-v})}{2 - v} = \frac{-6}{2 - v}$$

$$\Rightarrow \int (2 - v)\,dv = \int -\frac{6}{x}\,dx$$

$$\Rightarrow 2v - \frac{1}{2}v^2 = -6\ln|x| + C$$

$$\Rightarrow \boxed{2 \cdot \frac{y}{x} - \frac{1}{2}\left(\frac{y}{x}\right)^2 = -6\ln|x| + C}$$

implicit curves

Definition (Bernoulli Equations): A 1st-order equation of the form

$$\underbrace{\frac{dy}{dx} + P(x)y}_{\text{linear}} = Q(x)y^n \qquad (7.2)$$

nonlinear $n \neq 0, 1$
Bernoulli

where $P(x)$ and $Q(x)$ are continuous on an interval I and n is a real number is called a **Bernoulli Equation**.

NOTE: All linear equations are Bernoulli equations. In (7.2) if $n = 0$ or $n = 1$ then the equation is linear and can be solved using methods we have already outlined. Otherwise, when $n \neq 0, 1$ we make a substitution to convert the Bernoulli equation into a linear one.

$$\frac{1}{y^n}\left(\frac{dy}{dx} + P(x)y^1 = Q(x)\cancel{y^n}\right)$$

To see this, we divide both sides of (7.2) by y^n to obtain

$$y^{-n}\frac{dy}{dx} + P(x)y^{1-n} = Q(x) \qquad (7.3)$$

$y \to v$

$P(x)\,v$

If we wish for this to be a linear equation in variable v after some substitution, we will want the $P(x)y^{1-n}$ term to be $P(x)v$ instead. So we use the substitution $v = y^{1-n}$. Differentiating the substitution yields

$v = y^{1-n}$

$$\frac{dv}{dx} = (1-n)y^{(1-n)-1}\frac{dy}{dx} = (1-n)y^{-n}\frac{dy}{dx}$$

and by substituting this into the equation for $\frac{dy}{dx}$ and y^{1-n} we are left with

$$\Rightarrow \begin{cases} \dfrac{dy}{dx} = \dfrac{1}{1-n}\cdot y^n \dfrac{dv}{dx} \\ v = y^{1-n} \end{cases}$$

$$\cancel{y^{-n}}\left(\frac{1}{1-n}\cancel{y^n}\frac{dv}{dx}\right) + P(x)v = Q(x)$$

multiply by $1-n$: $\boxed{\dfrac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)}$

memorize

n power of y, $v = y^{1-n}$

Example: Solve

$$\underbrace{\frac{dy}{dx} + \frac{1}{x}y}_{\text{linear}} = x^2y^2$$

Bernoulli

$n = 2$, $P(x) = \frac{1}{x}$, $Q(x) = x^2$

substitution $\underline{v} = y^{1-n} = y^{1-2} = \underline{y^{-1}}$

$$\Rightarrow \frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)$$

$$\Rightarrow \frac{dv}{dx} + (1-2)\cdot\frac{1}{x}v = (1-2)(x^2)$$

$$\Rightarrow \frac{dv}{dx} \underbrace{- \frac{1}{x}}v = -x^2 \qquad \text{linear in } x, v$$

$$\mu(x) = e^{\int \tilde{P}(x)dx} = e^{\int -\frac{1}{x}dx} = e^{-\ln|x|} = x^{-1} = \frac{1}{x}$$

$$\Rightarrow \int \frac{d}{dx}\left(\frac{1}{x}v\right) = \frac{1}{x}\cdot(-x^2) = \int -x$$

$$\Rightarrow x\left(\frac{1}{x}v = -\frac{1}{2}x^2 + C\right)$$

$$\Rightarrow v = -\frac{1}{2}x^3 + Cx$$

revert $\Rightarrow$ $y^{-1} = -\frac{1}{2}x^3 + Cx$

$$\Rightarrow \boxed{y = \frac{1}{-\frac{1}{2}x^3 + Cx}}$$

Example: Solve

$$\underbrace{\frac{dy}{dx} + y}_{\text{linear}} = \underbrace{xy^{-2}}_{\text{Bernoulli}}$$

$n = -2, \quad P(x) = 1, \quad Q(x) = x$

use: $v = y^{1-n} = y^{1-(-2)} = y^3$

$$\frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)$$

$$\Rightarrow \frac{dv}{dx} + 3 \cdot 1 \cdot v = 3 \cdot x$$

$$\Rightarrow \frac{dv}{dx} + 3v = 3x \quad \text{(linear)}$$

$$\mu(x) = e^{\int 3dx} = e^{3x}$$

$$\Rightarrow \int \frac{d}{dx}(e^{3x}v) = \int e^{3x} \cdot 3x$$

IBP

$3x$	$+$	e^{3x}
3	$-$	$\frac{1}{3}e^{3x}$
0		$\frac{1}{9}e^{3x}$

$$\Rightarrow e^{3x}v = \int 3xe^{3x}dx + C$$

$$\Rightarrow e^{3x}v = 3x\left(\tfrac{1}{3}e^{3x}\right) - 3 \cdot \tfrac{1}{9}e^{3x} + C$$

$$\Rightarrow e^{3x}v = xe^{3x} - \tfrac{1}{3}e^{3x} + C$$

$$\Rightarrow v = x - \tfrac{1}{3} + Ce^{-3x}$$

$$\Rightarrow y^3 = x - \tfrac{1}{3} + Ce^{-3x} \Rightarrow \boxed{y = \sqrt[3]{x - \tfrac{1}{3} + Ce^{-3x}}}$$

Finally, if we are given a 1st-order equation of the form

$$\frac{dy}{dx} = G(\underline{ax + by + c})$$

we can make the substitution $\underline{z = ax + by + c}$ to transform the equation into one that is separable.

$$\frac{dz}{dx} = a + b\frac{dy}{dx}$$

Example: Solve

$$\frac{dy}{dx} = (\underbrace{x + y + 2}_{z})^2$$

Use $z = x + y + 2 \Rightarrow \frac{dz}{dx} = 1 + \frac{dy}{dx} + 0$

$$\Rightarrow \frac{dy}{dx} = \frac{dz}{dx} - 1$$

So : $\underbrace{\frac{dz}{dx} - 1}_{\frac{dy}{dx}} = z^2 \Rightarrow \frac{dz}{dx} = z^2 + 1$ (separable)

$$\Rightarrow \int \frac{1}{z^2+1}\,dz = \int dx$$

$$\Rightarrow \tan^{-1}(z) = x + C$$

$$\Rightarrow z = \tan(x + C)$$

$$\Rightarrow x + y + 2 = \tan(x + C)$$

$$\Rightarrow \boxed{y = \tan(x + C) - x - 2}$$